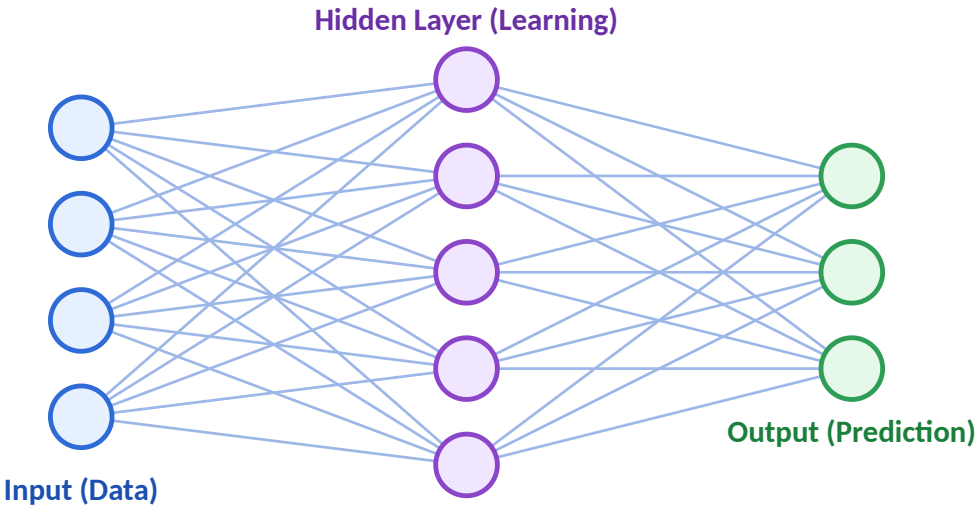
Complete Study Notes

Artificial Intelligence & Machine Learning

Concepts, diagrams, tables and revision questions – all in one place



What's inside

2 Artificial Intelligence & Machine Learning

3 Goals, Applications & Relationship

4 Types of Artificial Intelligence

5 Components & Differences

6 Types of Machine Learning

7 ML Workflow, Algorithms & Metrics

8 Data Splitting, Overfitting & Underfitting

9 Applications, Advantages & Disadvantages

10 Real World Examples & Future

11 AI vs ML vs DL vs Data Science

12 Careers, Tools & Frameworks

13 How AI & ML Work

14 Supervised & Unsupervised Learning

15 Reinforcement Learning

16 Project Workflow & Challenges

17 Summary, Terms & Revision Questions

Created by Tanmay Dwivedi

Artificial Intelligence & Machine Learning

1 What is Artificial Intelligence (AI)?

Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require **human intelligence**, such as learning, reasoning, problem-solving, understanding language and recognizing patterns.

Example : A voice assistant can understand a user's speech, process the request and provide a suitable response.

2 What is Machine Learning (ML)?

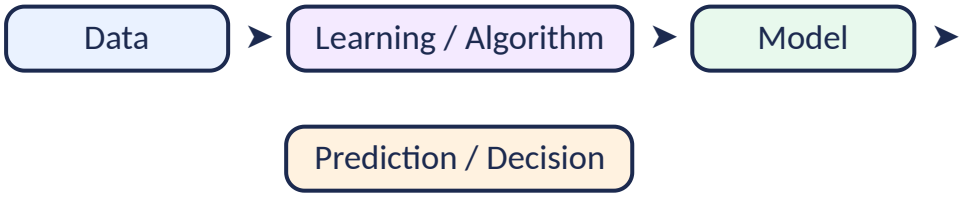
Machine Learning is a subset of AI in which computers learn patterns from data and improve their performance **without being explicitly programmed** for every task.

Example : An email spam filter learns from previous emails and predicts whether a new email is spam or not.

3 AI vs ML

AI	ML
Broader concept of making machines intelligent.	Technique used to achieve AI by learning from data.

4 Key Idea



★ Quick Facts

- The term **Artificial Intelligence** was coined by **John McCarthy** in 1956 (Dartmouth Conference).
- The term **Machine Learning** was popularized by **Arthur Samuel** in 1959.
- **Memory trick :** AI = the **goal**, ML = the **method**, DL = the **technique**.

Created by Tanmay Dwivedi

Goals, Applications & Relationship

5 Goals of Artificial Intelligence

The main goals of AI are :

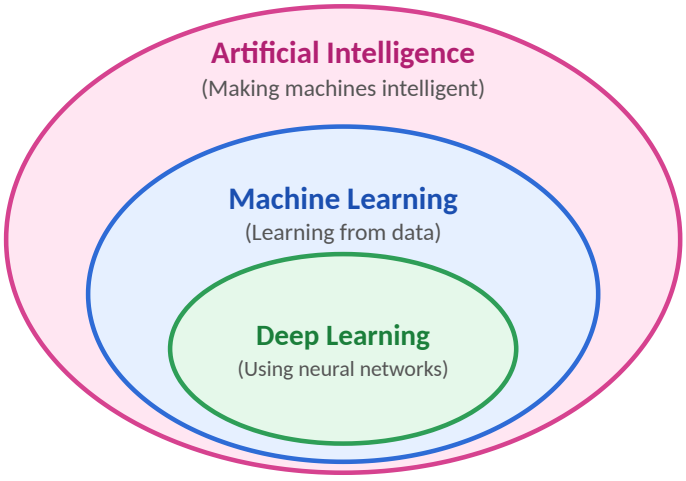
1. To create systems that can perform tasks which normally require human intelligence.
2. To enable machines to **learn from experience**.
3. To solve complex real-world problems.
4. To make machines able to understand, reason, plan and make decisions.
5. To improve efficiency and accuracy in various fields like healthcare, education, business, etc.

6 Applications of AI

AI is used in many real-world applications, such as :

- Virtual Assistants (Siri, Alexa, Google Assistant)
- Image Recognition (Face Unlock, Photo Tagging)
- Recommendation Systems (YouTube, Netflix, Amazon)
- Self-driving Cars
- Healthcare (Disease Prediction)
- Chatbots (Customer Support)
- Fraud Detection (Banking)
- Smart Home Devices

7 Relationship between AI, ML and Deep Learning



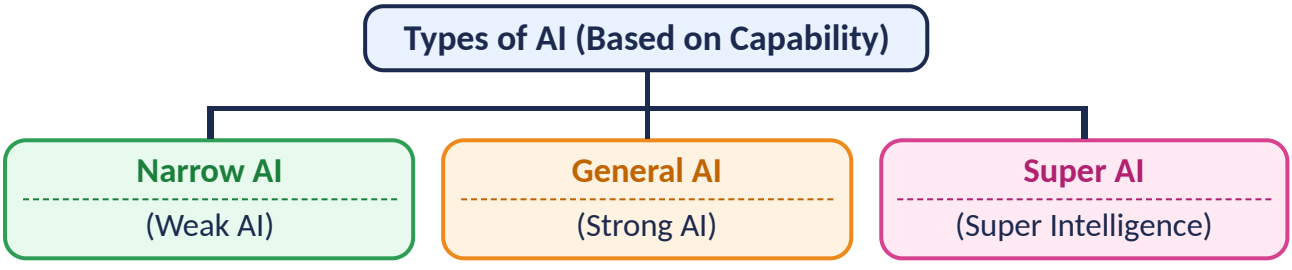
- **AI :**
  Broad field that enables machines to act intelligently.
- **ML :**
  Subset of AI that learns from data without explicit programming.
- **DL :**
  Subset of ML that uses multi-layer neural networks to learn patterns.

Types of Artificial Intelligence

8 Types of AI

AI can be classified in different ways. The two common ways are based on **capability** and based on **functionality**.

A. Based on Capability



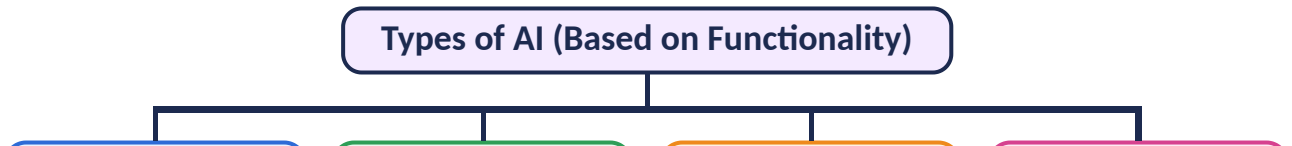
- Performs a specific task only
- Cannot think beyond trained domain

Ex: Siri, Google Assistant, Spam Filter

- Can learn and perform any intellectual task like a human
- Still under research

- Much smarter than humans
- May surpass human abilities
- Only a future concept

B. Based on Functionality



- Reacts to current input
- No memory of past

Ex: IBM Deep Blue (chess)

- Uses past data for decisions

Ex: Self-driving cars, recommendation systems

- Understands human emotions & beliefs

(Research stage)

- Has its own consciousness

(Hypothetical)

★ Easy Way to Remember

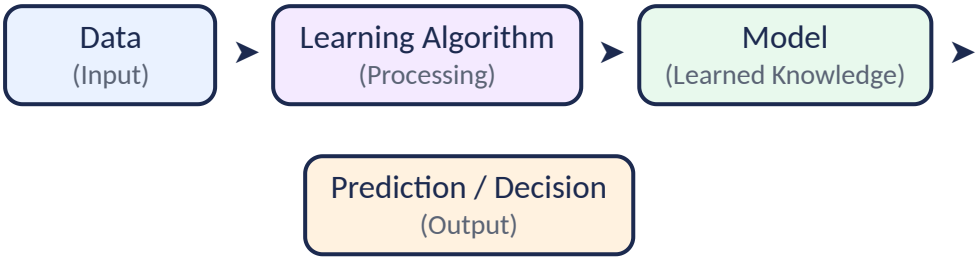
Reactive (no memory) ➤ Limited Memory (uses past data) ➤ Theory of Mind (understands emotions) ➤ Self-Aware (has consciousness)

Intelligence level increases from left to right. **All the AI we use today** (Siri, Netflix, ChatGPT, Google Maps) is **Narrow AI**.

Components & Differences

9 Components of an AI System

An AI system works with multiple components that work together to solve a problem.

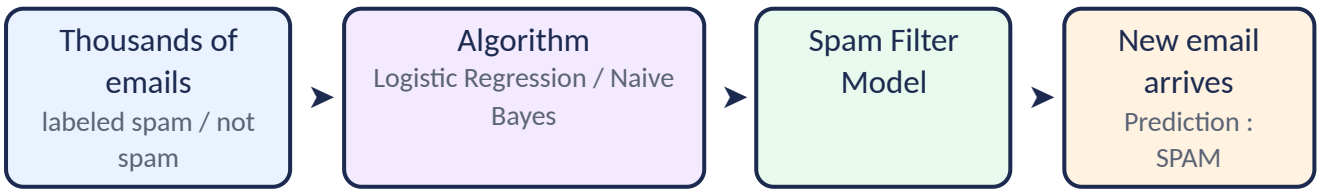


1. **Data :** Raw facts, numbers, text, images, etc. used for training the model.
2. **Learning Algorithm :** A method that learns patterns from data. Examples : Linear Regression, Decision Tree, Neural Network.
3. **Model :** A trained system that stores the learned patterns.
4. **Prediction / Decision :** The output generated by the model for new input data. Example : predicting house price, detecting spam email.

10 Difference between AI, ML and Deep Learning

Basis	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Meaning	Broad field of making machines intelligent	Subset of AI that learns from data	Subset of ML that uses neural networks with many layers
Data Requirement	Can work with or without data	Requires structured data	Requires large amount of data
Approach	Rule-based + learning	Learning from data	Learning with deep neural networks
Example	Expert systems, robotics	Spam detection, recommendation systems	Image recognition, speech recognition

★ Example : Spam Email Detection

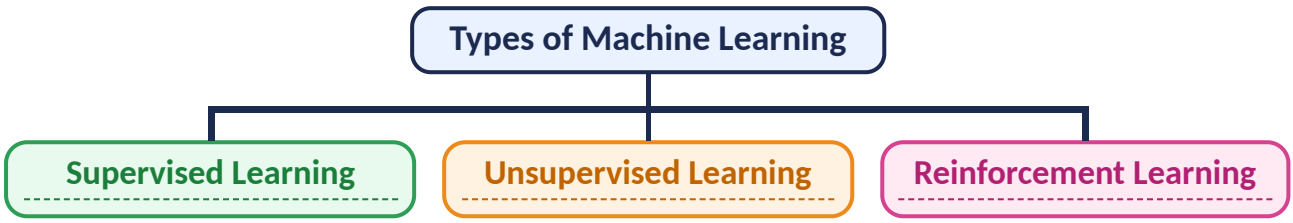


Note : The more good-quality emails the model sees, the better its predictions become.

Types of Machine Learning

11 Types of Machine Learning

Machine Learning can be mainly classified into three types based on the type of data and learning approach.



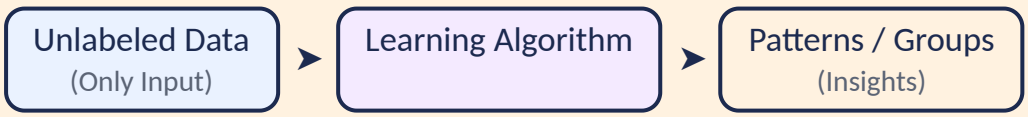
1. Supervised Learning

- Learns from **labeled data** (Input with correct output).
- Model learns a mapping between input and output.
- Used for classification and regression problems.
- Example : Email spam detection, house price prediction.



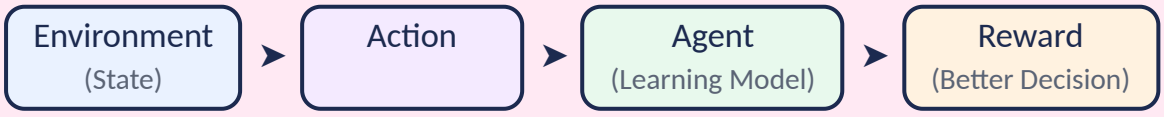
2. Unsupervised Learning

- Learns from **unlabeled data** (only input, no output).
- Finds hidden patterns or structures in data.
- Used for clustering and dimensionality reduction.
- Example : Customer segmentation, grouping similar news articles.



3. Reinforcement Learning

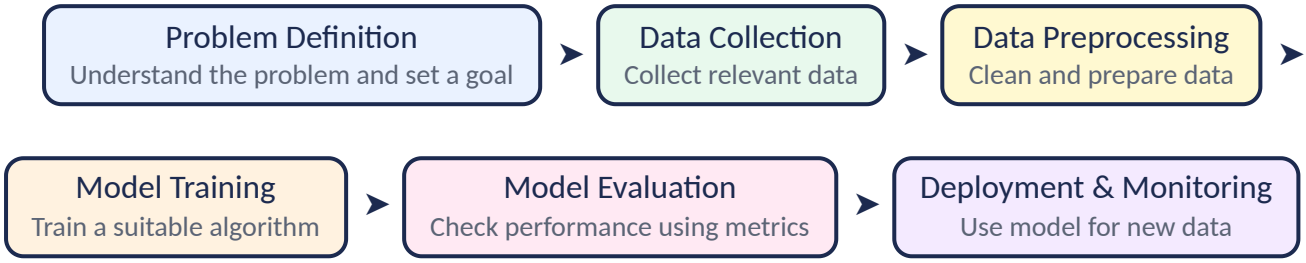
- Learns by **interacting with an environment**.
- Learns from rewards and penalties.
- Goal is to maximize the cumulative reward.
- Example : Game playing, robotics, self-driving cars.



ML Workflow, Algorithms & Metrics

12 Machine Learning Workflow

A typical ML project follows a sequence of steps from data collection to model deployment.



13 Common Machine Learning Algorithms

Algorithm	Type	Use Case
Linear Regression	Supervised (Regression)	Predict continuous values (price, sales)
Logistic Regression	Supervised (Classification)	Predict categorical values (spam / not spam)
Decision Tree	Supervised	Classification and regression
Random Forest	Supervised (Ensemble)	Higher accuracy, reduces overfitting
K-Means	Unsupervised (Clustering)	Customer segmentation
K-Nearest Neighbors	Supervised	Classification based on nearest data points
Support Vector Machine	Supervised	Text & image classification with maximum margin
Neural Networks	Supervised / Deep Learning	Image and speech recognition

14 Evaluation Metrics in Machine Learning

The performance of a model is measured using different metrics.

Type of Problem	Common Metrics	Meaning
Classification	Accuracy, Precision, Recall, F1-Score, Confusion Matrix	Measures how correctly the model predicts classes
Regression	MAE, MSE, RMSE, R^2	Measures the difference between predicted and actual values
Clustering	Silhouette Score, Davies-Bouldin Index	Measures how well the data is grouped

Data Splitting, Overfitting & Underfitting

15 Training, Validation and Testing Data

The dataset is usually divided into three parts to train and evaluate the model properly.

Original Dataset

Training Set (60% – 80%)	Validation Set (10% – 20%)	Testing Set (20% – 30%)
Used to train the model. Model learns patterns from the data.	Used to tune hyper-parameters and select the best model. Helps avoid overfitting.	Used to evaluate the final performance on unseen data.

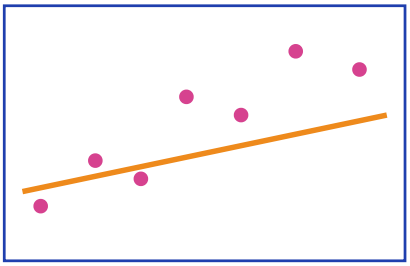
16 Overfitting and Underfitting

1. Underfitting

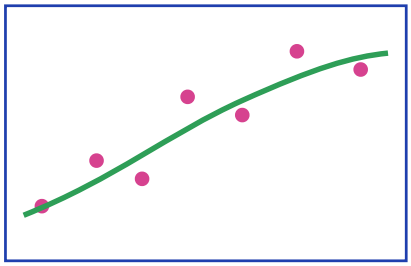
- Model is too simple to learn the underlying pattern.
- High bias, low variance.
- Performs poorly on both training and testing data.
- Example : a straight line for complex data.

2. Overfitting

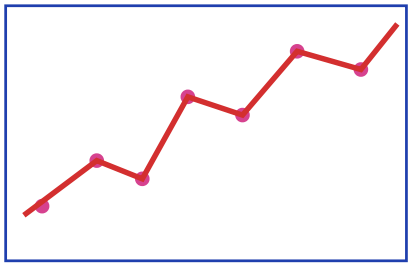
- Model learns training data too well, including noise.
- Low bias, high variance.
- Performs well on training data but poorly on testing data.
- Example : a very wiggly curve.



Underfitting (too simple)



Good Fit (just right)



Overfitting (too complex)

Aspect	Underfitting	Good Fit	Overfitting
Model Complexity	Too simple	Balanced	Too complex
Training Error	High	Low	Very low
Testing Error	High	Low	High
Bias	High	Low	Low
Variance	Low	Low	High

Applications, Advantages & Disadvantages

17 Applications of AI and Machine Learning

AI and ML are used in many real-world applications across various domains.

Healthcare
- Disease prediction
- Medical image analysis
- Drug discovery

Finance
- Fraud detection
- Credit scoring
- Stock market prediction

Education
- Personalized learning
- Automated grading
- Learning recommendations

E-commerce
- Product recommendation
- Customer segmentation
- Demand forecasting

Transportation
- Self-driving cars
- Traffic prediction
- Route optimization

Social Media
- Content recommendation
- Fake news detection
- Image and video analysis

Cybersecurity
- Threat detection
- Anomaly detection
- Malware classification

Agriculture
- Crop yield prediction
- Soil analysis
- Smart irrigation

18 Advantages and Disadvantages of AI / ML

Advantages
1. Can process large amounts of data.
2. Can find patterns and make accurate predictions.
3. Reduces human effort and saves time.
4. Works in dangerous or complex environments.
5. Improves decision-making.
6. Continuous improvement with more data.

Disadvantages
1. Requires large and high-quality data.
2. Can be costly and resource intensive.
3. May produce biased or unfair results if data is biased.
4. Lack of transparency (black-box models).
5. Can lead to job displacement.
6. Ethical and privacy concerns.

★ Principles of Responsible AI

Fairness	Transparency	Privacy	Accountability	Safety
No bias against any group	Decisions can be explained	User data is protected	Someone is responsible	Works reliably & safely

Real World Examples & Future

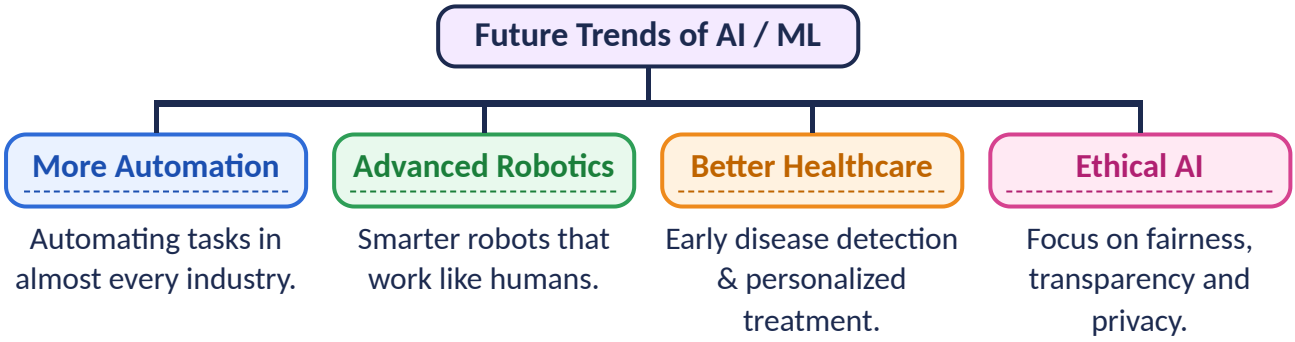
19 Real World Examples of AI and ML

AI and ML are already part of our daily life. Here are some popular real-world examples :

Application	Example
Voice Assistants	Siri, Google Assistant, Alexa
Recommendation Systems	YouTube, Netflix, Amazon
Image Recognition	Face Unlock, Google Photos
Spam Detection	Gmail
Self-driving Cars	Tesla, Waymo
Fraud Detection	Banks and payment apps
Language Translation	Google Translate
Virtual Chatbots	ChatGPT, Customer Support Bots

20 Future of AI and Machine Learning

AI and ML are rapidly evolving and will continue to grow. Some possible future trends :



★ Trending : Generative AI

> **Generative AI** is a modern branch of AI that can **create new content** – text, images, audio, video and code – after learning patterns from huge amounts of data.

Text
Chatbots, writing and summarizing

Images
Image generation and editing

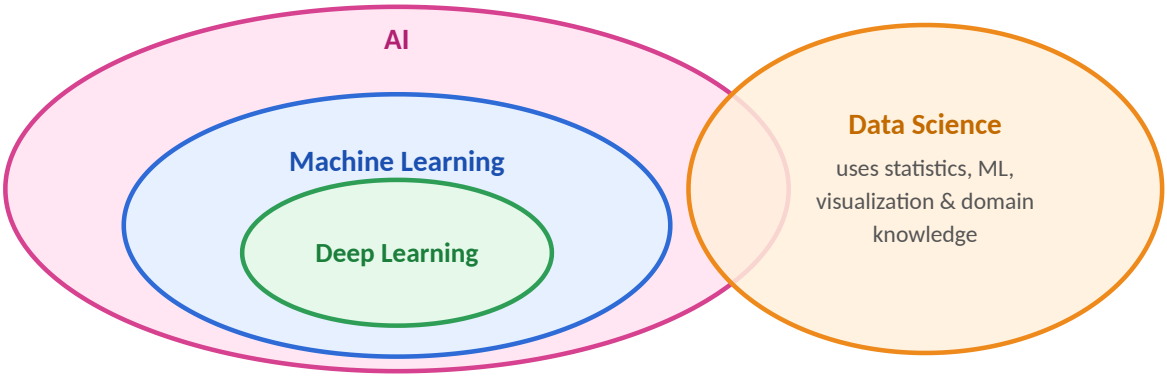
Code
Code suggestions and debugging

AI vs ML vs DL vs Data Science

21 Comparison

Aspect	AI	Machine Learning	Deep Learning	Data Science
Definition	Broad field of making machines intelligent	Subset of AI that learns from data	Subset of ML using deep neural networks	Field that extracts insights from data
Scope	Very wide	Narrower than AI	Narrower than ML	Focus on analysis & insight
Data Needed	Varies	Medium to large	Very large	Any size
Methods	Rules, search, learning	Regression, classification, clustering	CNN, RNN, LSTM, Transformers	Statistics, visualization, ML
Example	Chatbots, robots	Spam detection, recommendations	Image & speech recognition	Sales analysis, dashboards

22 How They Are Related



Key Points from this chapter

- AI is the broad field; ML is a subset of AI; DL is a subset of ML.
- ML learns from data and improves with experience.
- Data Science uses ML and other tools to get insights from data.

Careers, Tools & Frameworks

23 Career Opportunities in AI and ML

AI and ML have created a huge demand for skilled professionals across various industries.

Job Role	Key Responsibilities	Required Skills
Machine Learning Engineer	Build and deploy ML models in real applications	Python, ML algorithms, TensorFlow
Data Scientist	Analyze data and find meaningful insights	Statistics, Python / R, data visualization
AI Research Scientist	Work on new AI algorithms and techniques	Mathematics, deep learning, research skills
Data Analyst	Clean, analyze and visualize data	SQL, Excel, data visualization
NLP Engineer	Develop applications like chatbots, translators	NLP, deep learning, Python
Computer Vision Engineer	Work on image and video analysis	OpenCV, deep learning, image processing
AI Product Manager	Manage AI-based products and teams	Business understanding, AI knowledge

24 Tools and Frameworks for AI / ML

Python
- Most popular language for AI / ML
- Simple and easy to learn

NumPy
- Used for numerical computation
- Works with arrays and matrices

Pandas
- Data manipulation and analysis
- Easy to work with tabular data

Matplotlib
- Data visualization
- Helps to create graphs and charts

Scikit-learn
- Simple and efficient ML library
- Great for classification, regression, clustering

TensorFlow
- Developed by Google
- Used for deep learning

Keras
- High-level API for neural networks
- Easy to build and train deep learning models

PyTorch
- Developed by Meta (Facebook)
- Popular in research and deep learning

How AI & ML Work

25 How Machine Learning Works

Machine Learning works by learning from data. The general process of a machine learning model involves several steps :

1. Collect Data	Gather relevant data (images, text, numbers, etc.).
▼	
2. Preprocess Data	Clean the data, handle missing values, and convert it into a suitable format.
▼	
3. Split Data	Divide the data into training and testing sets.
▼	
4. Train the Model	Use algorithms to learn patterns from the training data.
▼	
5. Evaluate the Model	Test the model on unseen data to check its performance (accuracy, precision, etc.).
▼	
6. Make Predictions	Use the trained model to predict or make decisions on new data.

26 Types of Machine Learning – Comparison

Feature	Supervised	Unsupervised	Reinforcement
Definition	Learns from labeled data (input + output)	Learns from unlabeled data (only input)	Learns by interacting with an environment
Goal	Predict the correct output	Find hidden patterns	Maximize reward
Examples	Classification, regression	Clustering, dimensionality reduction	Game playing, robotics
Algorithms	Linear Regression, Decision Tree, SVM	K-Means, PCA	Q-Learning, Deep Q-Network
Use Cases	Spam detection, price prediction	Customer segmentation	Self-driving, game AI

★ Golden Rule : "Garbage In, Garbage Out"

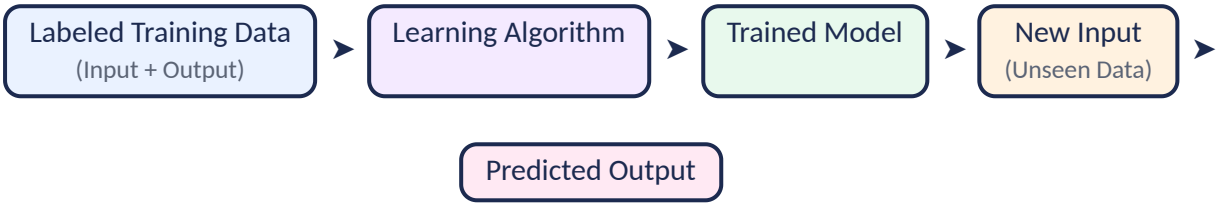
The quality of a model depends on the quality of its data. **Bad or biased data ⇒ bad or biased predictions**, however good the algorithm is.

Supervised & Unsupervised Learning

27 Supervised Learning

In supervised learning the model learns from **labeled data**, where both input and the correct output are provided. The model learns a mapping from inputs to outputs and can then predict the output for new, unseen data.

Working of Supervised Learning



1. Classification

- Predicts discrete or categorical output (class labels).
- Output is a category or class.
- Example : Spam or Not Spam, Disease Detection (Yes / No).

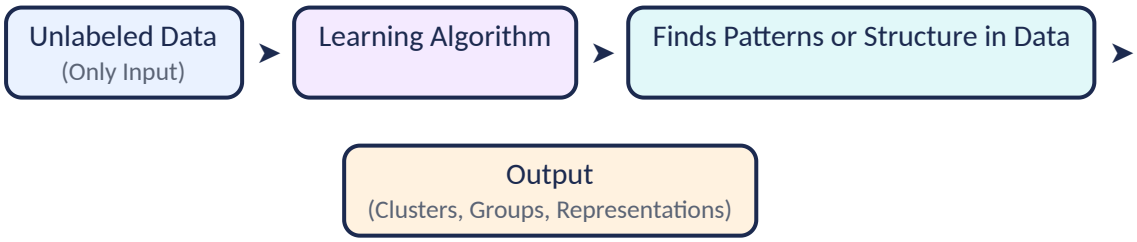
2. Regression

- Predicts continuous numerical output.
- Output is a real value.
- Example : House Price Prediction, Temperature Prediction.

28 Unsupervised Learning

In unsupervised learning the model learns from **unlabeled data**, where only input data is provided. The model tries to find hidden patterns, similarities or structure in the data without any predefined output.

Working of Unsupervised Learning



1. Clustering

- Groups data into similar clusters.
- Example : Customer segmentation, document clustering.

2. Dimensionality Reduction

- Reduces the number of features while preserving important information.
- Example : PCA (Principal Component Analysis).

3. Association Rule Learning

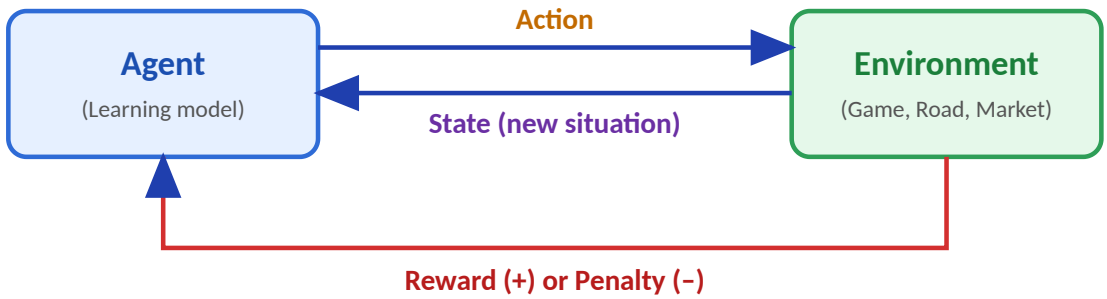
- Finds relationships between items.
- Example : Market basket analysis (customers who buy bread also buy butter).

Reinforcement Learning

29 Reinforcement Learning

Reinforcement Learning is a type of machine learning in which an **agent** learns by interacting with an **environment**. The agent takes actions, receives rewards or penalties, and learns a policy to maximize the cumulative reward over time.

Working of Reinforcement Learning



Component	Description
Agent	Learner or decision maker
Environment	Everything the agent interacts with
State	Current situation of the agent
Action	Move made by the agent
Reward	Feedback for an action (positive or negative)
Policy	Strategy the agent follows to choose actions
Goal	Maximize the total cumulative reward

30 Applications of AI and ML in Industries

Domain	Applications	Examples
Healthcare	Disease prediction, medical image analysis	Cancer detection, patient risk prediction
Finance	Fraud detection, credit scoring, algorithmic trading	Detecting fake transactions
E-commerce	Recommendation systems, customer segmentation	Amazon, Flipkart suggestions
Transportation	Self-driving, route and traffic prediction	Tesla Autopilot, Google Maps
Education	Personalized learning, automated grading	Adaptive learning platforms
Cybersecurity	Threat detection, anomaly detection	Phishing and malware detection
Entertainment	Content recommendation, content generation	YouTube, Netflix

Project Workflow & Challenges

31 AI / ML Project Workflow

Building an AI or ML project follows a systematic workflow from problem understanding to deployment.

1. Problem Definition	Understand the objective; define input, output and success criteria.
▼	
2. Data Collection	Gather relevant data from databases, APIs, sensors, web, etc.
▼	
3. Data Preprocessing	Clean data, handle missing values, remove duplicates, perform feature engineering.
▼	
4. Exploratory Data Analysis (EDA)	Visualize and analyze data; find patterns and relationships.
▼	
5. Model Selection	Choose the appropriate ML / DL algorithm; split data into train and test sets.
▼	
6. Model Training & Evaluation	Train on training data, tune hyper-parameters, evaluate on unseen data.
▼	
7. Deployment & Monitoring	Deploy the model to real-world use and keep monitoring its performance.

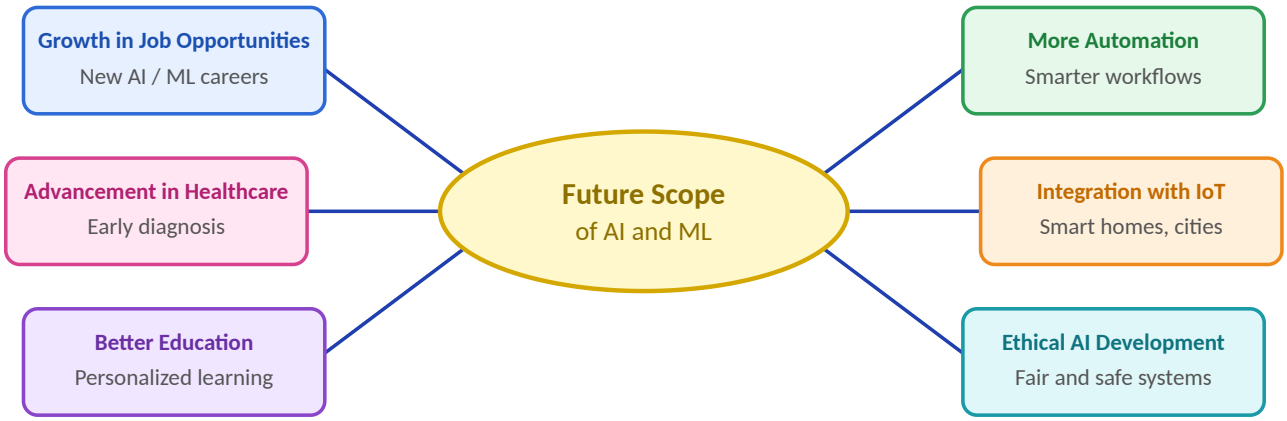
32 Challenges in AI and Machine Learning

AI and ML have great potential but also face several challenges :

Challenge	Description
Data Quality	Poor quality or biased data can lead to wrong predictions.
Data Privacy	Handling sensitive data raises privacy and security concerns.
Computational Cost	Training large models needs high computational resources and time.
Bias and Fairness	Models may inherit unfair bias from training data.
Explainability	Many models are a "black box" and hard to interpret.
Ethical Issues	Misuse of AI, job displacement, accountability.
Lack of Skilled People	High demand for AI / ML experts.

Summary, Terms & Revision Questions

33 Future Scope of AI and ML



34 Important Terms to Know

Term	Meaning
Dataset	Collection of data used for training and testing
Feature	Input variable or attribute of the data
Label	Output or target variable
Model	Learned representation from data
Training	Process of learning from data
Testing	Checking model performance on unseen data
Accuracy	Percentage of correct predictions
Overfitting	Model learns training data too well, fails on new data
Underfitting	Model is too simple to learn the pattern

35 Practice / Revision Questions

Q. No.	Question
1	What is Artificial Intelligence? Explain with example.
2	What is Machine Learning? How is it different from AI?
3	What are the types of Machine Learning? Explain each with an example.
4	What is Deep Learning? How is it different from Machine Learning?
5	Give some real-world applications of AI and ML.
6	What are the advantages and disadvantages of AI?
7	What is overfitting and underfitting? How can they be reduced?
8	Explain the workflow of an ML project.
9	What is the difference between classification and regression?
10	What is the future scope of AI and Machine Learning?

— All the best for your revision ✍️